

User Manual

RMP PLUS 401

Safety

Improper use of the RMP PLUS 401 can cause personal injury, death and/or property damage from loss of control, collision, and falls. To reduce risk

of injury, read and follow all instructions and warnings in this manual.

The following safety messaging conventions are used throughout this document:

WARNING!	Warns you about actions that could result in death or serious injury.
CAUTION!	Warns you about actions that could result in minor or moderate injury.
NOTICE	Indicates information considered important, but not related to personal injury.

WARNING!

- Do not sit, stand or ride on RMP PLUS 401. It may cause injury.
- Do not control RMP PLUS 401 to hit people or animals. Collision may cause injury.
- When RMP PLUS 401 is running, it needs to remind people nearby at all times. Accidental collisions with RMP PLUS 401 may result in injuries.

- RMP PLUS 401 can be accelerated quickly, and customers are advised to practice at a low speed until users are familiar with controlling RMP PLUS 401. The unexpected movement of RMP PLUS 401 may cause injuries.
- Do not attempt to disassemble the battery, which may result in electric shock, burns or even fire. Trying to open the battery case can damage the battery case and release toxic substances. It can also render the battery unusable.
- As with all rechargeable batteries, do not charge near flammable materials, which could cause a fire.
- If the battery shell is damaged or the battery emits peculiar smell, smoke, overheat, or leaks, do not continue to use the battery. Do not touch any substance that oozes from the battery, which may cause poisoning.
- Strictly observe and follow all safety information on the warning label on the battery. Failure to do so can result in injury or even death.
- Do not use badly worn or damaged cables, which may shock yourself or damage RMP PLUS 401.

CAUTION!

- Set performance parameters correctly and carefully. RMP PLUS 401

follows the commands issued to it, and it is the user's responsibility to implement correct and safe performance parameters.

- Not charging the battery can cause permanent damage to the battery.
- The battery can only be charged with the charger of RMP PLUS 401.
- Before operating RMP PLUS 401, please be sure to read the user manual and be familiar with the operation of RMP PLUS 401 and various precautions.

NOTICE

- In the absence of communication with our company, our company will not take any responsibility for any accident caused by the modification of the chassis.

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1 Product introduction

1.1 Product schematic diagram

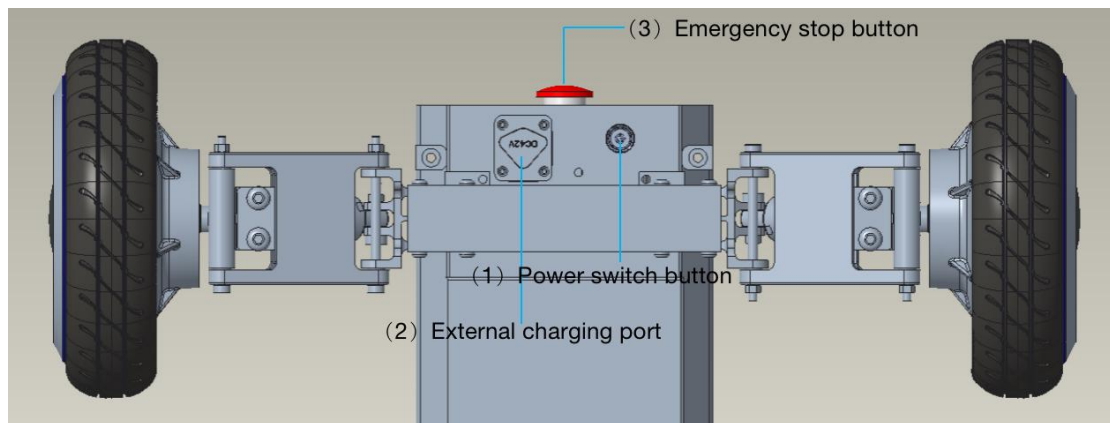


Figure 1

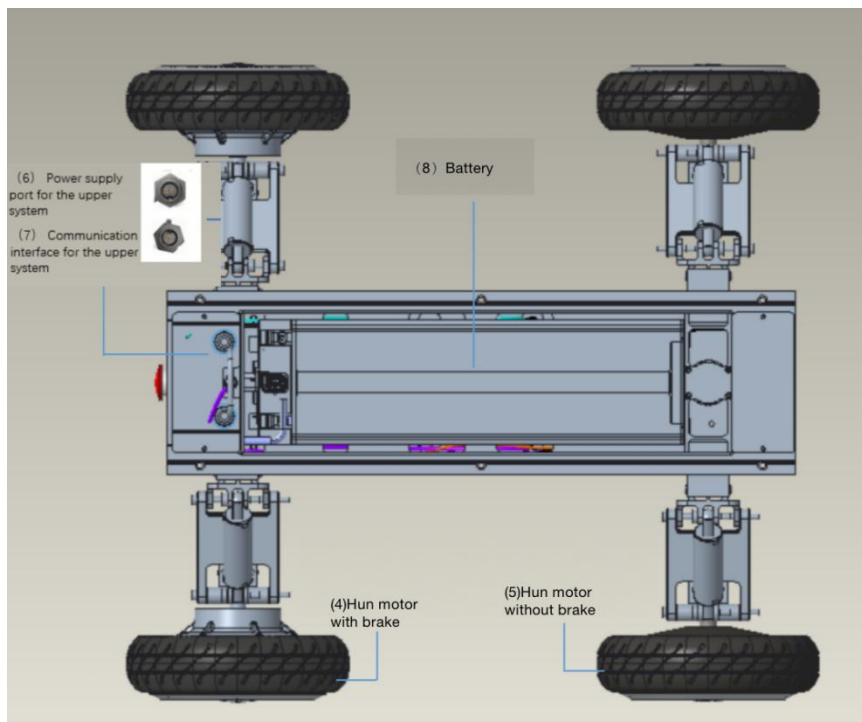


Figure 2

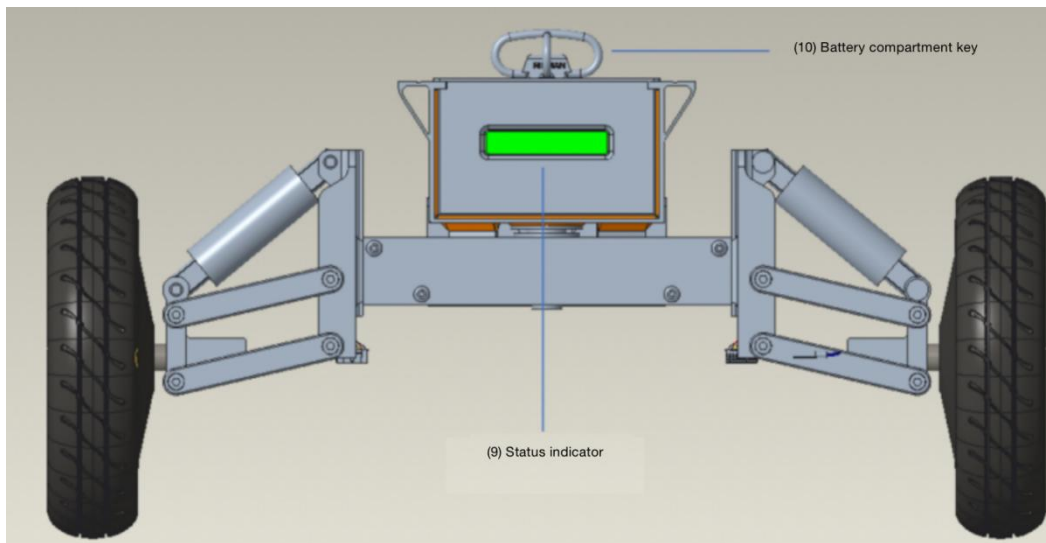


Figure 3

1.2 Component Introduction

Table 1

Number	Component name	Description
1	Power switch button	Start up: long press the power switch button, until the indicator light is steady on and accompanied by a prompt tone, the chassis starts up successfully, the chassis is in lock mode, and the indicator light is steady yellow Shutdown: long press the power switch button until the prompt sound starts, release the power switch button, the chassis shuts down successfully, at this time, the power switch button lamp and indicator light are all off.
2	External charging port	Connect the charger to charge the device
3	Emergency stop button	Used to switch the chassis to emergency stop mode in emergency state
4	Hub motor with brake	8 inch tire, with power-off brake
5	Hub motor without brake	8 inch tire
6	Power supply port for the upper system	Supplies power to the upper system (Voltage: 36V) You need to use DCDC module to convert the voltage to match your upper system
7	Communication interface for the upper system	Includes CAN, serial port, and remote control receiver
8	Battery	Used to power the entire system
9	Status indicator	Indicator color and status represent different modes of the product

10	Battery compartment key	It opens the lock that holds the battery
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1.3 The remote control

1.3.1 Schematic diagram of remote control



Figure 4

* The forward and reverse input (throttle or direction) of the remote control can be achieved by flipping the phase switch under the T8FB.



Figure 5

*Remote control alarm voltage adaptive 2S, 3S, 4S lithium batteries and 4 nickel metal hydride batteries, that is, if T8FB is powered by 2S, 3S, 4S lithium batteries or 4 nickel metal hydride batteries, after connecting the batteries, T8FB will automatically set the low voltage alarm value according to the battery type.

1.3.2 Receiver pair

Each transmitter has its own ID code. Before starting to use the device, the receiver must pair with the transmitter. After pairing, the ID code is stored in the receiver and does not need to be paired again unless the receiver is paired with another transmitter. When you buy a new receiver, you must re-code it, otherwise the receiver will not work properly.

(1) Place the remote control and receiver horizontally, and the distance between them is about 50cm;

(2) Turn on the power switch of the remote control to supply power to the receiver, and the LED light of the receiver starts to blink slowly;

(3) Press the ID SET key on the side of the receiver for more than 1 second, and the LED light of the receiver starts flashing quickly, indicating that the code is being matched, and the receiver will look for the nearest remote control for matching the code;

(4) Stop flashing the LED light of the receiver, which means that the code is completed. If the LED light of the receiver blinks slowly, it means that the code fails, and the code needs to be repeated.

1.3.3 Remote control instructions

(1) RMP PLUS 401 chassis startup: press the RMP PLUS 401 power switch button;

Note: Please check the RMP PLUS 401 status. Hold down the power switch button until the buzzer does not beep continuously and the indicator is steady yellow.

(2) Turn on the remote control: push up the power switch of the remote control;

Note: Ensure that the remote control is not in emergency stop state and enter the enable state. That is, the emergency stop switch is not under, and the enable switch is shifted from the top to the bottom.

(3) RMP PLUS 401 is in Normal mode at this time. See the following table for detailed operations:

Table 2 Remote control operation

Control action	Remote control operation
Turn right or left	Turn the rudder lever left and right
Move forward or back	The throttle lever moves forward and backward
Emergency stop/exit emergency stop	Remote control emergency stop switch: Top - exit emergency stop; Bottom - Open emergency stop
Adjust the maximum angular velocity	Adjustment knob for maximum angular velocity of remote control. If you turn left, the maximum angular velocity decreases. If you turn right, the maximum angular velocity increases.
Adjust the maximum linear velocity	Adjustment knob for maximum linear speed of remote controller. If you turn left, the maximum linear velocity decreases. If you turn right, the maximum linear velocity increases.
Enable/disable the function	Turn the enable switch from the top to the bottom to enable chassis. Turn the enable switch from the bottom to the top to disable the chassis.

1.3.4 The upper system controls the car explanation

The upper system is a PC terminal control computer, which can

directly issue control commands, and various information changes are displayed on the screen. The upper system controls the chassis and provides some necessary operating environment for the chassis, and extends the man-machine control or demonstration functions provided by the chassis. The upper system has the characteristics of leading management, coordinating resources, monitoring agent and controlling RMP PLUS 401.

(1) RMP PLUS 401 chassis startup: press the RMP PLUS 401 power switch button;

Note one: Please check the RMP PLUS 401 status. Hold down the power switch button until the buzzer does not beep continuously and the indicator is steady yellow.

Note two: The remote control cannot be turned on when the upper system controls the car. Or after the remote control is turned on, turn the enable switch to the top.

(2) Ensure that the RMP PLUS 401 serial cable or CAN line is connected to the upper system;

(3) Grant permission to the /sdcard/segway/hardware_log/ folder. Otherwise, the new log file will fail. Give /catkin_ws/SRC/RosCode segwayRMP PLUS 401/lib/ all files and directory add permissions (after the first setting, you do not need to reset it) :

```
‘ ’  
  
`cd /sdcard/segway/hardware_log`  
  
`sudo chmod 777 /sdcard/segway/hardware_log/`  
  
`cd $PRO_HOME$/catkin_ws/src/RosCode/segwayRMP PLUS  
401/lib/`  
  
`sudo chmod 777 *`  
  
‘ ’
```

(4) According to the file of txt under ‘catkin_ws/src/RosCode /segwayRMP PLUS 401/Cmakelists.txt’ , choose the compile option based on the upper system platform (x86_64 or arm). The sample below is compiled based on x86_64 platform. Disable ‘libctrl_arm64-v8a.so’ by using ‘#’ (after the first setting, you do not need to reset it) :

```
‘ ’  
  
`target_link_libraries(SmartCar`  
  
`${catkin_LIBRARIES}`  
  
`#${PROJECT_SOURCE_DIR}/lib/libctrl_arm64-v8a.so  
  
//under x86_64 platform, enable the line, disable the line under ARM`  
  
`${PROJECT_SOURCE_DIR}/lib/libctrl_x86_64.so  
  
// under arm platform, enable the line, disable the line under x86_64`  
  
‘ ’
```

(5) Enter the ROS workspace and run the following command to

compile the Segway_MSgs package message.

```
‘ ’
```

```
cd catkin_ws
```

```
catkin_make
```

```
-DCATKIN_WHITELIST_PACKAGES='segway_msgs'
```

```
‘ ’
```

(6) Enter the ROS workspace and run the following command to compile the Segwayrmp package.

```
‘ ’
```

```
cd catkin_ws
```

```
catkin_make -DCATKIN_WHITELIST_PACKAGES='segwayrmp'
```

```
‘ ’
```

(7) Execution of vehicle control in ROS:

1) To create a terminal, run the following command

```
‘ ’
```

```
cd catkin_ws
```

```
roscore
```

```
‘ ’
```

2) Create a new terminal and run the SmarCar node

```
‘ ’
```

```
cd catkin_ws
```

```
source devel/setup.bash

roslaunch segwayrmp SmartCar
```

‘ ’

3) Create a new terminal and run the following command to run the routine test node

‘ ’

```
cd catkin_ws

source devel/setup.bash

roslaunch segwayrmp ChassisResponseTest
```

‘ ’

2 Software introduction

This chapter describes the related files, software interface functions, and fault code information provided by RMP PLUS 401.

2.1 The file provided to the user

Table 3 Files provided

Files	Function
Libctrl_x86_64.so	Provides the x86 platform C/C++ chassis related interface
Libctrl_arm64-v8a.so	Provides arm platform C/C++ chassis related interface

Comm_ctrl_navigation.h	The C/C++ API interface head files
ROS package	Provides ROS nodes for chassis control

2.2 Interface function introduction

2.2.1 C/C++interface introduction

Table 4 callback data type

The callback type	Index of the callback	Function description	Data structure
Chassis_Data_Motors_Speed	1	Chassis 4 wheel speed information	<pre>typedef struct{ int16_t fl_speed; int16_t fr_speed; int16_t rl_speed; int16_t rr_speed; }chassis_motors_speed_data_t;</pre>
Chassis_Data_Car_Speed	2	Get chassis speed information	<pre>typedef struct{ int16_t car_speed; int16_t turn_speed; }chassis_car_speed_data_t;</pre>
Chassis_Data_Front_Ticks	3	Chassis front two wheel encoder information	<pre>typedef struct{ int32_t fl_ticks; int32_t fr_ticks; }front_motors_ticks_t;</pre>
Chassis_Data_Rear_Ticks	4	Chassis rear two wheel encoder information	<pre>typedef struct{ int32_t rl_ticks; int32_t rr_ticks; }rear_motors_ticks_t;</pre>
Chassis_Data_Odom_Pose_xy	5	Odom pose information	<pre>typedef struct{ float pos_x; float pos_y; }odom_pos_xy_t;</pre>
Chassis_Data_Odom_Euler_xy	6	Odom Euler X/Y information	<pre>typedef struct{ float euler_x; float euler_y; }odom_euler_xy_t;</pre>

Chassis_Data_Odom_Euler_z	7	Odom Euler Z information	typedef struct{ float euler_z; }odom_euler_z_t;
Chassis_Data_Odom_Linevel_xy	8	Odom speed X/Y information	typedef struct{ float vel_line_x; float vel_line_y; }odom_vel_line_xy_t;
Chassis_Data_Imu_Gyr	9	Gyroscope data	typedef struct{ int16_t gyr[3]; }imu_gyr_original_data_;
Chassis_Data_Imu_Acc	10	Accelerometer data	typedef struct{ int16_t acc[3]; }imu_acc_original_data_;

Note 1: Odom data: the default heading Angle at startup is 0 degrees.

Note 2: IMU (gyroscope and accelerometer) data: carrier coordinate system XYZ corresponds to right, front and up.

Table 5 event definition

Event type	Index of event	Function description
ChassisBootReadyEvent	1	The chassis center panel is started
PadPowerOffEvent	2	The chassis to turn it off
OnEmergeStopEvent	3	Enter the emergency stop
OutEmergeStopEvent	4	Exit the emergency stop
OnLockedRotorProtectEvent	5	The inability of a wheel to rotate due to an external force
OutLockedRotorProtectEvent	6	The failure of the wheel to rotate due to external forces was eliminated
OnLostCtrlProtectEvent	7	The wheel appears to rotate rapidly without control
OutLostCtrlProtectEvent	8	The rapid rotation phenomenon is eliminated
CalibrateGyroSuccess	9	Gyroscope calibration successful
CalibrateGyroFail	10	Failed to calibrate the gyroscope
CalibratePasheCurrentSuccess	11	Calibration of phase current

		succeeded
CalibratePasheCurrentFail	12	Failed to calibrate phase current
ChassisLockRotorWarning	13	Locked-rotor occur and then warning

Table 6 get/set interface

Interface name	Interface description
get_err_state	Get error code for upper system/central board/motor board/battery
get_bat_soc	Obtain the percentage of the remaining battery power
get_bat_charging	Obtain the battery charging status (1: charging; 0: non-charging)
get_bat_mvolt	Obtain battery voltage (unit: millivolt)
get_bat_mcurrent	Obtain battery current (unit: milliampere)
get_bat_temp	Obtain the battery temperature (in Degrees Celsius)
get_chassis_work_model	Get chassis working state (0: wheels have no power; 1: the wheel have power)
get_chassis_load_state	Get chassis load parameter Settings (0: no load; 1: full load)
get_chassis_mode	Get the chassis state machine (0: lock the car; 1: car control; 2. Implementation; 3: emergency stop; 4: False)
get_ctrl_cmd_src	Get the chassis current control source (0: remote control; 1: host computer)
get_vehicle_meter	Get chassis mileage (unit meters)
get_host_version	Obtain the version number of the upper computer
get_chassis_central_version	Obtains the version number of the central board
get_chassis_motor_version	Get motor board version number (reserved)
get_line_forward_max_vel_fb	Obtain chassis forward speed limit value (unit meters per hour)
get_line_backward_max_vel_fb	Get chassis backward speed limit value (unit meters per hour)
get_angular_max_vel_fb	Get the chassis angular velocity limit (milliradians per second)
getIapProgress	Get IAP progress
iapCentralBoard	IAP upgrades to the central board
iapMotorBoard	IAP upgrades to the motor board
iapBrakeBoard	IAP upgrades to the brake board(with the brake board)
isHostIapOver	Check whether IAP ends
getHostIapResult	Get IAP results (3: complete; 4: failure. 5: Interruption. 0: meaningless)
getHostIapErrorCode	Get the IAP error code
get_calibrate_mid_value_status	Check whether the front wheel median has been corrected (1: corrected; 0: uncorrected)
set_cmd_vel	Set chassis linear velocity and angular velocity (unit meters per

	second and radians per second)
set_line_forward_max_vel	Set chassis forward speed limit (in meters per second)
set_line_backward_max_vel	Set chassis backward speed limit (in meters per second)
set_angular_max_vel	Set chassis angular velocity limit (unit radians per second)
set_enable_ctrl	Set the enable state of the chassis upper system controlled vehicle (1: Enable; 0: Disable)
init_control_ctrl	Chassis initialization interface
exit_control_ctrl	Chassis exit initialization interface
set_smart_car_serial	Set the name of the serial port used by the dynamic library of the upper computer
set_comu_interface	Set the communication interface with the chassis (0: serial port; 1: CAN)
set_chassis_load_state	Set chassis load parameters (0: no load; 1: full load)
set_chassis_poweroff	Issue the chassis shutdown command
setHostIapCanceled	Cancel IAP command on upper computer
set_calibrate_mid_value	The command to correct the median Angle of the first two rounds is delivered
reset_host_power_time_s	Issue a command to reset the upper mechanical power of the chassis (unit: second; Maximum interval: 65535 seconds)
clear_chassis_error_code	Clear chassis error codes, excluding alarms, exceptions, and battery errors. Use with caution
enable_rotate_switch	Enable the chassis in-situ rotation function

2.2.2 ROS interface introduction —SmartCar

Table 7 News release

Topic Name	Function description	Message Type	Message Type Info	Frequency
Bms_fb	Battery Information	Segway_msgs/ Bms_fb	int16 bat_soc int16 bat_charging int32 bat_vol int32 bat_current int16 bat_temp	1
Chassis_ctrl_src_fb	Chassis control command source	Segway_msgs/ Chassis_ctrl_src_fb	uint16 chassis_ctrl_cmd_src	1
Chassis_mileage_meter_fb	Chassis mileage	Segway_msgs/ Chassis_mileage_meter_fb	uint32 vehicle_meters	1
Chassis_mode_fb	Chassis state machine	Segway_msgs/ Chassis_mode_fb	uint16 chassis_mode	1
Error_code_fb	Chassis error code	Segway_msgs/ Error_code_fb	uint32 host_error uint32 central_error uint32 front_left_motor_error uint32 front_right_motor_error uint32 rear_left_motor_error uint32 rear_right_motor_error uint32 bms_error	1
Motor_work_mode_fb	Chassis working condition	Segway_msgs/ Motor_work_mode_fb	uint16 motor_work_mode #0: no output torque 1: output torque	1
Speed_fb	Chassis speed	Segway_msgs/ Speed_fb	float32 car_speed float32 turn_speed float32 fl_speed float32 fr_speed float32 rl_speed float32 rr_speed uint64	40

			speed_timestamp	
Ticks_fb	Chassis encoder information	Segway_msgs/ Ticks_fb	int32 fl_ticks int32 fr_ticks int32 rl_ticks int32 rr_ticks uint64 ticks_timestamp	40
Odom	Odom data	Nav_msgs/odom		40
Imu	Imu data	Sensor_msgs/imu		40

Table 8 News subscription

TopicName	Function decription	Message Type	Message Type Info
Cmd_vel	Control chassis movement	Geometry_msgs/twist	Angular.z //rad/s Linear.x //m/s

Table 9 service client

Service name	Function decription	Message type	Message type info
chassis_send_event_srv	Send event number	Segway_msgs/chassis_send_event	chassis_send_event_id--ros_is_received

Table10 service server

Service name	Function decription	Message type	Message type info
ros_clear_chassis_error_code_cmd.srv	Clear chassis error codes, excluding alarms, exceptions, and battery errors. Use with caution	Segway_msgs/ ros_clear_chassis_error_code_cmd	bool clear_chassis_error_code_cmd --- uint8 clear_chassis_error_code_result
ros_enable_chassis_rotate_cmd.srv	Enable the chassis in-situ rotation	Segway_msgs/ ros_enable_chassis_rotate_cmd	bool ros_enable_chassis_rotate_cmd ---

	function		int16 chassis_enable_rotate_result
ros_get_chassis_rotate_switch_cmd_srv	Query whether the chassis is rotating in place	Segway_msgs/ ros_get_chassis_rotate_switch_cmd	bool ros_get_chassis_rotate_cmd --- uint8 chassis_rotate_state
ros_get_chassis_SN_cmd_srv	Get the SN of the chassis central control MCU	Segway_msgs/ ros_get_chassis_SN_cmd	bool ros_get_chassis_SN --- string chassis_SN
ros_get_load_param_cmd_srv	Getting load Settings	Segway_msgs/ ros_get_load_param_cmd	ros_get_load_param --- get_load_param #0:no_load, 1: full_load
ros_get_sw_version_cmd_srv	Obtaining the Software Version	Segway_msgs/ ros_get_sw_version_cmd	ros_get_sw_version_cmd --- uint16 host_version uint16 central_version uint16 motor_version
ros_get_vel_max_feedback_cmd_srv	Get the speed limiter	Segway_msgs/ ros_get_vel_max_feedback_cmd	ros_get_vel_max_fb_cmd --- forward_max_vel_fb backward_max_vel_fb angular_max_vel_fb
ros_set_chassis_enable_cmd_srv	The chassis enable command is issued	Segway_msgs/ ros_set_chassis_enable_cmd	ros_set_chassis_enable_cmd --- chassis_set_chassis_enable_result
ros_set_chassis_poweroff_cmd_srv	Issue the chassis shutdown command	Segway_msgs/ ros_set_chassis_poweroff_cmd	ros_set_chassis_poweroff_cmd --- chassis_set_poweroff_result
ros_set_load_param_cmd_srv	Setting chassis load	Segway_msgs/ ros_set_load_param_cmd	ros_set_load_param #0:no_load, 1: full_load --- chassis_set_load_param_result
ros_set_vel_max_cmd_srv	Set the speed limit	Segway_msgs/ ros_set_vel_max_cmd_srv	ros_set_forward_max_vel ros_set_backward_max_vel ros_set_angular_max_vel --- chassis_set_max_vel_result

ros_reset_host_power_cmd_srv	Chassis reset upper power	segway_msgs/ros_reset_host_power_cmd	uint16 reset_interval_time --- uint8 reset_result
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Table11 action server

Action name	Function decription	Message type	Message type info
ros_set_iap_cmd_action	IAP upgrade of board firmware	Segway_msgs/ros_set_iap_cmdAction	uint16 board_index_for_iap --- Int16 iap_result #3: iap_state_complete; 4: iap_state_fail; 5: iap_state_abort Int16 error_code #When iap_result value is 4, this value represents the error code --- Int16 iap_percent

2.2.3 Fault code information table

The fault code is obtained through the uint32_get_err_state (board_name_e board_name) interface. The corresponding information is as follows: (**Note: "Manual Force Clear Errors" needs to be implemented very carefully**) :

Table12 Fault code

Board name	Bit	Error info	底盘动作	处理
host	0x00000000	No error		
	0x00000001	The central board is out of contact	Level 2, unable to control the car	Check communication
	0x00000002	The serial port module is removed	Level 2, unable to control the car	Plug back the module

Central	0x00000000	No error		
	0x00000001	Vehicle control command communication is interrupted	Level 2, chassis parking, lock mode	Check communication
	0x00000002	Motor board communication is interrupted	Level 2, chassis parking, lock mode	Check motor board communication
	0x00000004	IMU initialization failed	Level 1, controllable car, wrong angle	Drive back at low speed, check hardware
	0x00000008	The IMU failed to read data	Level 1, controllable car, wrong angle	Drive back at low speed, check hardware
	0x00000010	The wheel whirled uncontrollably	Level 5, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware
	0x00000020	The wheel cannot rotate because of external forces	Level 3, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware
	0x00000040	Failed to calibrate the IMU	Level 1, controllable car, wrong angle	Drive back at low speed, check hardware
	0x00000080	Flash read failure	Level 3, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware
	0x00000100	IMU data update failed	Level 1, controllable car, wrong angle	Drive back at low speed, check hardware
	0x00000400	rollover	Level 2, Parking, Lock	Make sure the chassis is not in

			Mode	a rollover state
	0x00000800	Restart of any motor board detected	Level 2, Parking, Lock Mode	Check motor board communication
	0x00001000	The front left wheel magnetic encoder is faulty	Level 5, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware
	0x00002000	The front right wheel magnetic encoder is faulty	Level 5, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware
	0x00004000	Battery communication is interrupted.	Level 3, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware
	0x00008000	The rear left wheel magnetic encoder is faulty	Level 5, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware
	0x00010000	The rear right wheel magnetic encoder is faulty	Level 5, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware
	0x00200000	Abnormal front wheel angle convergence timeout	Level 2, Parking, Lock Mode	Righting the front wheel
Motor	0x00000000	No error		
	0x00000001	Phase current fault	Level 4, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware
	0x00000002	Phase voltage fault	Level 3, Parking, Error Mode	Manually clear errors, drive to safety at low

				speed, check hardware
0x00000004	Lack of phase	Level 3, Parking, Error Mode		Manually clear errors, drive to safety at low speed, check hardware
0x00000008	Voltage failure	Level 4, Parking, Error Mode		Manually clear errors, drive to safety at low speed, check hardware
0x00000010	Self-test failure	Level 3, Parking, Error Mode		Manually clear errors, drive to safety at low speed, check hardware
0x00000020	Over current	Level 4, Parking, Error Mode		Manually clear errors, drive to safety at low speed, check hardware
0x00000080	The wheel cannot rotate because of external forces	Level 4, Parking, Error Mode		Manually clear errors, drive to safety at low speed, check hardware
0x00000100	Electrical Angle fault	Level 3, Parking, Error Mode		Manually clear errors, drive to safety at low speed, check hardware
0x00000200	Overpower fault	Level 4, Parking, Error Mode		detect hardware
0x00000400	Over speed fault	Level 5, Parking, Error Mode		Manually clear errors, drive to safety at low speed, check hardware
0x00000800	Speed sensor fault	Level 3, Parking, Error		Manually clear errors, drive to

			Mode	safety at low speed, check hardware
Battery	0x00000000	No error	No error	
	0x00000200	overshoot	Level 3, Parking, Error Mode	detect hardware
	0x00000400	The charging temperature exceeds the normal temperature	Level 3, Parking, Error Mode	detect hardware
BrakeSticking	0x00000002	Left brake open circuit	Level 3, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware
	0x00000004	Left brake short circuit	Level 4, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware
	0x00000008	Left brake failed to close	Level 3, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware
	0x00000200	Right brake open circuit	Level 3, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware
	0x00000400	Right brake short circuit	Level 4, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware
	0x00000800	Right brake failed to close	Level 3, Parking, Error Mode	Manually clear errors, drive to safety at low speed, check hardware

	0x00008000	Brake board communication lost	Level 2, Parking, Error Mode	Detect brake board communication
--	------------	--------------------------------------	------------------------------------	--

2.2.4 In-situ rotation function introduction

The in-situ rotation function is a function realized in the special mode of the chassis. When the linear velocity is 0 and the angular velocity is not 0, the chassis uses this function. When this function is used, the current of the rear wheel will be too large, which may cause abnormality of the chassis and the motor. It is recommended to use this function. Not necessary and not easy to use.

When using the chassis to rotate in place, if the chassis is stuck and cannot rotate and the rotor is blocked, the chassis will perform the following actions:

1. When the rotor is locked for about 5 seconds, the chassis cancels the support for the in-situ rotation function, and sends the locked rotor alarm event to the upper computer at the same time. At this time, it is necessary to stop the rotation in place, and then let the chassis have a forward and backward movement component to release the electric energy accumulated by the chassis.

2. When the locked rotor occurs and the chassis still cannot release the electric energy for about 10 seconds, the chassis will automatically

release the accumulated electric energy. At this time, it is necessary to prevent the chassis from being on a sloped ground.

3. When the locked rotor occurs, the hardware is damaged and the power cannot be released. After about 15 seconds, the chassis sends a locked rotor error event to the upper computer. At the same time, the chassis switches to the error mode, and the power is off and the brake is locked.

After the chassis releases power and returns to normal, if it is really necessary to continue to use the in-situ rotation function, as described in Section 2.2.2, the host computer needs to use the "Enable chassis in-situ rotation mode" service of ROS to enable the chassis to support in-situ again. rotation. At the same time, it should be noted that in order to protect the chassis, the above services cannot be used within 30 seconds after the chassis stops supporting the in-situ rotation function to enable the chassis to support the in-situ rotation function.

3 Firmware upgrade and version upgrade

IAP is a software function module of the system, that is, in application programming, that is, to upgrade the single-chip program online. This function uses the host computer to program the new version of the bin file of the MCU (including the central control board,

motor drive board, etc.) when the program is running: for the release version before V2.02.3, the premise is that the MCU bin file to be name it according to the requirements of the host computer (such as *central.bin*), and then place it in the */sdcard/firmware/* path of the host computer; For V2.02.3 and later versions, you need to name the bin file of the MCU to be programmed according to the requirements of the host computer (such as *central_v2023.bin*), and then place it in the *segway_firmware* folder under the folder where the lib library and ctrl_x86_64 are located; after that Perform online upgrades through commands on the terminal.

3.1 The firmware update

Before firmware upgrade, test the data communication between the upper computer and the lower computer to check whether the communication is normal. Test using commands at shell terminals.

(1) View the program path of the upper computer

Go to the path where the host program resides and check whether the executable file exists. Arm executable file, x86 executable file, ARM dynamic library and x86 dynamic library are shown below:

```
ubuntu@ubuntu: /home/project/EE_PROJECT_RMP/Project/RMP_panda/ROS/src/segwayrmp/lib$ ll
总用量 1742
drwxrwxrwx 1 root root 4096 6月 30 10:11 /
drwxrwxrwx 1 root root 4096 6月 16 17:29 /
-rwxrwxrwx 1 root root 93115 4月 20 17:50 adb*
-rwxrwxrwx 1 root root 414192 6月 30 10:11 ctrl_arm64-v8a*
-rwxrwxrwx 1 root root 386280 6月 30 10:11 ctrl_x86_64*
-rwxrwxrwx 1 root root 446544 6月 30 10:11 libctrl_arm64-v8a.so*
-rwxrwxrwx 1 root root 433816 6月 30 10:11 libctrl_x86_64.so*
ubuntu@ubuntu: /home/project/EE_PROJECT_RMP/Project/RMP_panda/ROS/src/segwayrmp/lib$
```

Figure 6

(2) View the software version of central board

Check the software version of the lower computer. This step can test the communication status of the upper computer and the lower computer at the same time. If you can check the software version of each plate of the lower computer through the upper computer, it indicates that the communication is good.

Center board test command: `./ctrl_x86_64 s -test central`

1) When connecting the serial port for the first time, if the USB port of the serial port is not granted with the execution permission, the program requires to obtain the root permission to modify the execution permission of the serial port USB port. In this case, you need to enter the system login password and press Enter, as shown in the following figure:

```
ubuntu@ubuntu: /home/project/EE_PROJECT_RMP/Project/RMP_panda/ROS/src/segwayrmp/lib$ ./ctrl_x86_64 s -test central
.....Start Comucore!.....
host version build date:[21-06-29]
host version build time:[20:39:35]
Communication interface adding SERIAL_INTERFACE
Use the serial port[/dev/ttyUSB0]
Please enter the administrator permission login password:
[sudo] ubuntu 的密码:
```

Figure 7

2) When communication fails, the version number is 0xFFFF, as shown in the figure below:

```
ubuntu@ubuntu: /home/project/EE_PROJECT_RMP/Project/RMP_panda/ROS/src/segwayrmp/lib$ ./ctrl_x86_64 s -test central
.....Start Comucore!.....
host version build date:[21-06-29]
host version build time:[20:39:35]
Communication interface adding SERIAL_INTERFACE
Use the serial port[/dev/ttyUSB0]
Please enter the administrator permission login password:
[sudo] ubuntu 的密码:
serial open success! serial port:/dev/ttyUSB0, baud:921600
Scheduler Num 0 Start. Task Num = 1. Period = 100000
Scheduler Num 1 Start. Task Num = 1. Period = 50000
Scheduler Num 2 Start. Task Num = 1. Period = 20000
当前测试RMP版本: 1.0.0

central board test started.....
get_chassis_central_version: 0xFFFF
get_chassis_Err_Status: 0x00000000
get_chassis_central_version: 0xFFFF
get_chassis_Err_Status: 0x00000000
get_chassis_central_version: 0xFFFF
get_chassis_Err_Status: 0x00000000
```

Figure 8

3) When the communication is successful, the version number is printed as follows, and the version number is non-0xFFFF. At this time, the communication between the upper computer and the single chip microcomputer is normal, and online upgrade can be carried out:

```
ubuntu@ubuntu: /home/project/EE_PROJECT_RMP/Project/RMP_panda/ROS/src/segwayrmp/lib$ ./ctrl_x86_64 s -test central
.....Start Comucore!.....
host version build date:[21-06-29]
host version build time:[20:39:35]
Communication interface adding SERIAL_INTERFACE
Use the serial port[/dev/ttyUSB0]
Please enter the administrator permission login password:
serial open success! serial port:/dev/ttyUSB0, baud:921600
Scheduler Num 1 Start. Task Num = 1. Period = 50000
Scheduler Num 0 Start. Task Num = 1. Period = 100000
Scheduler Num 2 Start. Task Num = 1. Period = 20000
当前测试RMP版本: 1.0.0

central board test started.....
get_chassis_central_version: 0x1000
get_chassis_Err_Status: 0x00000000
get_chassis_central_version: 0x1000
get_chassis_Err_Status: 0x00000000
```

图 9

3.2 Version update

(1) Single-chip bin file placement

For releases before V2.02.3, put the bin file of the board software to be upgraded under the host computer/sdcard/firmware path, the central control board software bin file central.bin, the front wheel motor board bin file motor_front.bin, and the rear wheel The motor board bin file motor_rear.bin, the brake board bin file brake.bin (specific to the chassis with brake), as shown in the file location below

```
ubuntu@ubuntu:/sdcard/firmware$ ll
总用量 272
drwxrwxrwx 2 root root 4096 1月 7 10:33 ./
drwxrwxrwx 5 root root 4096 7月 26 16:28 ../
-rwxrwxrwx 1 root root 28708 11月 16 10:39 brake.bin*
-rwxrwxrwx 1 ubuntu ubuntu 91368 12月 31 14:59 central.bin*
-rwxrwxrwx 1 root root 67788 11月 12 20:49 motor_front.bin*
-rwxrwxrwx 1 root root 67788 11月 12 20:49 motor_rear.bin*
-rwxrwxrwx 1 root root 40 10月 13 10:09 password.txt*
ubuntu@ubuntu:/sdcard/firmware$
```

Figure 10a

For V2.02.3 and later releases, put the firmware bin file of the board to be upgraded into the segway_firmware folder under the folder where the upper computer libctrl_x86_64.so is located, such as the central control board bin file central_v2023.bin, and the front wheel motor board bin The file motor_front_v2023.bin, the rear wheel motor board bin file motor_rear_v2023.bin, the brake board bin file brake_v2023.bin (specific to the chassis with brake board), as shown in the following figure:

```
ubuntu@ubuntu:/home/project/git_pro/EE_PROJECT_RMP/Project/RMP_panda/ROS/src/segwayrmp/lib/segway_firmware$ ll
总用量 268
drwxrwxr-x 2 ubuntu ubuntu 4096 3月 30 15:25 ./
drwxrwxr-x 3 ubuntu ubuntu 4096 3月 30 15:25 ../
-rw-rw-r-- 1 ubuntu ubuntu 28952 3月 30 15:25 brake_v2024.bin
-rw-rw-r-- 1 ubuntu ubuntu 93516 3月 30 15:25 central_v2024.bin
-rw-rw-r-- 1 ubuntu ubuntu 68672 3月 30 15:25 motor_front_v2024.bin
-rw-rw-r-- 1 ubuntu ubuntu 68672 3月 30 15:25 motor_rear_v2024.bin
ubuntu@ubuntu:/home/project/git_pro/EE_PROJECT_RMP/Project/RMP_panda/ROS/src/segwayrmp/lib/segway_firmware$
```

Figure 10a

(2) The bin file on the lower computer is written online

Enter the path of the program executable file ctrl_X86_64 or

CTRL_ARM64-V8A on the upper computer, as follows:

```
ubuntu@ubuntu:/home/project/EE_PROJECT_RMP/Project/RMP_panda/ROS/src/segwayrmp/lib$ ll
总用量 1763
drwxrwxrwx 1 root root 4096 6月 30 10:11 /
drwxrwxrwx 1 root root 4096 6月 16 17:29 /
-rwxrwxrwx 1 root root 114264 6月 30 14:13 adb*
-rwxrwxrwx 1 root root 414192 6月 30 10:11 ctrl_arm64-v8a*
-rwxrwxrwx 1 root root 386280 6月 30 10:11 ctrl_x86_64*
-rwxrwxrwx 1 root root 446544 6月 30 10:11 libctrl_arm64-v8a.so*
-rwxrwxrwx 1 root root 433816 6月 30 10:11 libctrl_x86_64.so*
ubuntu@ubuntu:/home/project/EE_PROJECT_RMP/Project/RMP_panda/ROS/src/segwayrmp/lib$
```

Figure 11

Run the following commands to upgrade each board online: Go to the program path of the upper computer and run the following commands (use 's' when using a serial port; Use 'C' when using CAN port) :

Center board upgrade command:

./ctrl_arm64-v8a c -iap central

Front wheel motor board upgrade command:

./ctrl_arm64-v8a c -iap motor_front

Rear wheel motor board upgrade command :

./ctrl_arm64-v8a c -iap motor_rear

Brake board upgrade command :

`./ctrl_arm64-v8a c -iap brake`

For example, run the `./ ctrl_x86_64 s -iap central` command to upgrade the central board, as shown in the following figure:

```
ubuntu@ubuntu:/home/project/EE_PROJECT_RMP/Project/RMP_panda/ROS/src/segwayrmp/lib$ ./ctrl_x86_64 s -iap central
.....Start Comucore!.....
host version build date:[21-06-29]
host version build time:[20:39:35]
Communication interface adding SERIAL_INTERFACE
Use the serial port[/dev/ttyUSB0]
Please enter the administrator permission login password:
serial open success! serial port:/dev/ttyUSB0, baud:921600
Scheduler Num 0 Start. Task Num = 1. Period = 100000
Scheduler Num 1 Start. Task Num = 1. Period = 50000
Scheduler Num 2 Start. Task Num = 1. Period = 20000
当前测试RMP版本: 1.0.0

IAP Start! path:/sdcard/firmware/central.bin id: 38 version:2.01
Id:0x38 version:2.01 Iap Progress 0: status: 2
Id:0x38 version:2.01 Iap Progress 0: status: 2
Id:0x38 version:2.01 Iap Progress 0: status: 2
Id:0x38 version:2.01 Iap Progress 0: status: 2
Id:0x38 version:2.01 Iap Progress 0: status: 2
Id:0x38 version:2.01 Iap Progress 0: status: 2
Id:0x38 version:2.01 Iap Progress 0: status: 2
Id:0x38 version:2.01 Iap Progress 0: status: 2
Id:0x38 version:2.01 Iap Progress 0: status: 2
Id:0x38 version:2.01 Iap Progress 0: status: 4
Id:0x38 version:2.01 Iap Progress 0: status: 4
Id:0x38 version:2.01 Iap Progress 0: status: 4
Id:0x38 version:2.01 Iap Progress 0: status: 4
Id:0x38 version:2.01 Iap Progress 1: status: 4
Id:0x38 version:2.01 Iap Progress 1: status: 4
Id:0x38 version:2.01 Iap Progress 1: status: 4
Id:0x38 version:2.01 Iap Progress 1: status: 4
Id:0x38 version:2.01 Iap Progress 2: status: 4
```

Figure 12

During the upgrade process, you can view the upgrade Progress. Progress indicates the percentage of IAP upgrade Progress. When the Progress value reaches 100, it indicates that the bin file of the

routing board has been burned to the central board chip.

```
Id:0x38 version:2.01 Iap Progress 97: status: 4
Id:0x38 version:2.01 Iap Progress 98: status: 4
Id:0x38 version:2.01 Iap Progress 98: status: 4
Id:0x38 version:2.01 Iap Progress 98: status: 4
Id:0x38 version:2.01 Iap Progress 98: status: 4
Id:0x38 version:2.01 Iap Progress 99: status: 4
Id:0x38 version:2.01 Iap Progress 99: status: 4
Id:0x38 version:2.01 Iap Progress 99: status: 4
Id:0x38 version:2.01 Iap Progress 99: status: 4
Id:0x38 version:2.01 Iap Progress 99: status: 8
Id:0x38 version:2.01 Iap Progress 99: status: 8
Id:0x38 version:2.01 Iap Progress 99: status: 8
Id:0x38 version:2.01 Iap Progress 99: status: 8
Id:0x38 version:2.01 Iap Progress 99: status: 8
Id:0x38 version:2.01 Iap Progress 99: status: 8
Id:0x38 version:2.01 Iap Progress 99: status: 8
Id:0x38 version:2.01 Iap Progress 99: status: 8
Id:0x38 version:2.01 Iap Progress 100: status:10
Iap_success!
ubuntu@ubuntu:/home/project/EE_PROJECT_RMP/Project/RMP_panda/ROS/src/segwayrmp/lib$
```

Figure 13

(3) Test the results of the IAP version online upgrade:

Run Step 1 to check the software version. In the path of the upper computer program, enter the command `./ctrl_x86_64 s -test central`, as shown in the following figure:

In this case, the software version number of the central control board is 0x1000, indicating that the online upgrade is successful and the communication between the host computer and the central control board is good.

```
ubuntu@ubuntu:/home/project/EE_PROJECT_RMP/Project/RMP_panda/ROS/src/segwayrmp/lib$ ./ctrl_x86_64 s -test central
.....Start Comucore!.....
host version build date:[21-06-29]
host version build time:[20:39:35]
Communication interface adding SERIAL_INTERFACE
Use the serial port[/dev/ttyUSB0]
Please enter the administrator permission login password:
serial open success! serial port:/dev/ttyUSB0, baud:921600
Scheduler Num 1 Start. Task Num = 1. Period = 50000
Scheduler Num 0 Start. Task Num = 1. Period = 100000
Scheduler Num 2 Start. Task Num = 1. Period = 20000
当前测试RMP版本: 1.0.0

central board test started.....
get_chassis_central_version: 0x1000
get_chassis_Err_Status: 0x00000000
get chassis central version: 0x1000
```

Figure 14

Appendix 1 System Parameters and Mode switching logic

Table 1 System parameters

The structural parameters	size	Length * Width * Height (mm) 672 * 617 * 274
	The structural parameters	Wheelbase * wheel base * ground clearance (mm) : 456*545*58 (chassis compressed to the lowest point)
	Tire size	8.5 inch
	own weight	28kg
	Nominal load	28kg
	The disabled suspension	5cm/10° ramp/speed bump Independent suspension
	Protection grade	IPX5
The performance parameters	Maximum speed	3.56m/s
	Maximum steering speed	2rad/s
	Minimum turning radius	1.36m
	The braking distance	Under full load, 3.56m/s about 1m
	The control mode	Remote control, upper computer control
	The brake way	The electric brake 、 mechanical brake
Communication	Communication interface	UART, CAN
	API	ROS
	Feedback data	encoder, IMU
The battery	Continuous operating distance	under full load 2m/s 5 、 about 40km
	The battery	36V 15.3Ah
	Charging way	Manual cable charge/quick battery change
Interaction	The keys	Emergency stop botton, power swich botton
	Status indication	On/off state indicator light, chassis state indicator light, Control source indicator, power display, charging state display

Table 2 Mode switching logic

Chassis mode	input	excute	exit
Lock mode	1. The default mode for powering on the chassis 2. Default mode after emergency stop is restored 3. In the vehicle control mode, it enters the mode after recoverable exceptions such as communication timeout and communication chain disconnection occur 4. Enter this mode after the upper computer manually forcibly clears the error	Version without brake: 0-speed closed-loop, shielding speed command, the status indicator light is always on in yellow; Version with brake: Shield the speed command, the status indicator light is always on in yellow, the brake is disabled when power off, and the brake is locked.	1. An unrecoverable exception errorcode is detected and the system enters the error mode 2. The enable command is received and the vehicle control mode is entered 3. The emergency stop button is triggered to enter the emergency stop mode
Control mode	In the lock mode, the enable command is received	Close loop, accept control instruction. Remote control car - green indicator light is often blinking; Upper control car - Indicator is green on (The version with a power-off brake will open the brake when the brake is powered on)	1. An unrecoverable exception errorCode is detected and the system enters the error mode 2. Enter the lock mode after detecting recoverable exceptions such as communication timeout and communication chain disconnection 3. The emergency stop button is triggered to enter the emergency stop mode
Emergency stop mode	In non-abnormal mode, the emergency stop button is triggered	Wheels lose power, shielding speed and	The emergency stop button is up and the lock mode is entered

		enable command, status indicator is red and often flashes.(The version with the power-off brake is disabled at this time, and the brake is locked)	
Error mode	An unrecoverable exception errorcode was detected	Brake, wheels lose power shielding speed and enable command. Indicator light steady red (The version with the power-off brake is disabled at this time, and the brake is locked)	1. Reboot 2. The upper computer manually forcibly clears the error. (manual forced clearing of errors needs to be implemented very carefully)

Appendix 2 Connector welding instructions

一、 The preparatory work

1、 Tools

Electric iron, solder wire

2、 The material

8 Pin connector, 2 Pin connector, two AWG16 cables, and eight AWG26 cables, as shown in Figure 1.

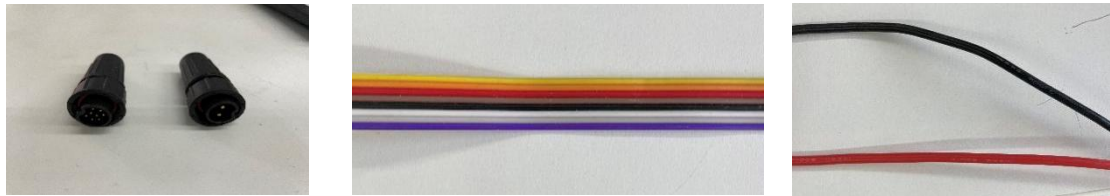


Figure 1

二、 Welding instructions (taking 8 Pin connector as an example)

1. The 8 Pin connector received by the customer is shown in Figure 2.

Screw the connector from the position shown in the red arrow to the state shown in Figure 3;



Figure 2



Figure 3

2. Take out the component shown in Figure 4, which is the component to

be welded;



Figure 4

3. As shown in Figure 5, the Pin Angle number of the connector can be seen from one side of the part. Then rotate the part 180°, which is the part to be welded;



Figure 5

4. Weld the wires of AWG26 according to the definition of PIN Angle in the welding instructions (see Appendix 3 for details). After the welding is completed, see Figure 6;

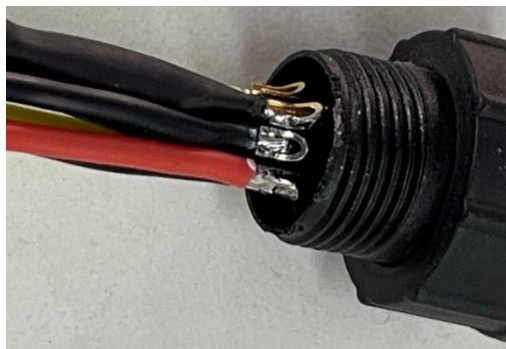


Figure 6

5. Take out the two parts as shown in Figure 7 and put them on the welded parts, as shown in Figure 8;

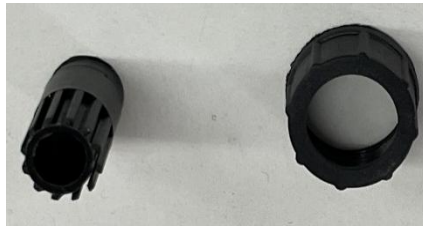


Figure 7

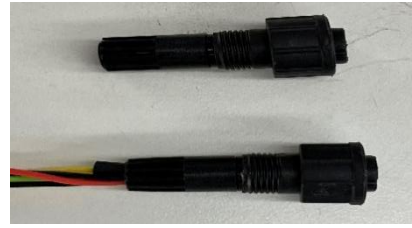


Figure 8

6. Take out the parts shown in Figure 9, put them on the previously assembled parts, and tighten them, as shown in Figure 10;



Figure 9



Figure 10

7. Then connect the remote control receiver and serial port, as shown in Figure 11;

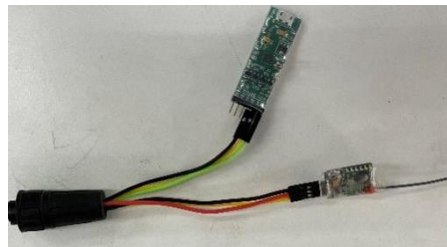


Figure 11

8. The welding method of the 2 Pin connector is the same as that of the 8 Pin connector.

Appendix 3 Defines the connector Pin Angle

Connector	Pin number	Define	Wire size	Remark	Color
8pin	1	CANH	AWG26	CAN	Red
	2	CANL	AWG26		Gray
	3	TX	AWG26	Serial port	Blue
	4	RX	AWG26		Green
	5	GND	AWG26		White
	6	5V	AWG26	Remote control receiver	Brown
	7	GND	AWG26		Black
	8	S.B PPM	AWG26		Yellow
2pin	1	Power+	AWG16	power supply for upper system	Red
	2	Power-	AWG16		Black

Appendix 4 C/C++ API reference documentation

➤ `int Init_Comcore(void)`

Function: initialization of host computer dynamic link library

Parameter: none

Return value: 0: initialization succeed;

Other: initialization fail

➤ `void exit_Comcore(void)`

Function: exit initialization of host computer dynamic link library

Parameter: none

Return value: none

➤ `void aprctrl_datastamped_jni_register(saprctrl_datastamped* f)`

Function: registration via the callback function provided by parameter, and this callback function conducts the sensor data processing.

Parameter: f is a struct pointer, and this struct includes the unique function pointer member variables.

Return value: none

➤ `void aprctrl_eventcallback_jni_register(saprctrl_eventt* f)`

Function: registration via the callback function provided by parameter, and this callback function conducts the processing of event code.

Parameter: f is a struct pointer, and this struct includes the unique function pointer member variables.

Return value: none

➤ `uint16_t get_err_state(boardname_e boardname)`

Function: acquire the software/firmware runtime error code

Parameter: board name refers to the software/firmware ID

Parameter is one of the following values:

Host upper computer node ID

Motor0 front_left motor node ID

Motor1 front_right motor node ID

Motor2 rear_left motor node ID

Motor3 rear_right motor node ID

Central central_board node ID

BMS batter ID

Return value: error code

➤ `int16_t get_bat_soc(void)`

Function: acquire percentage of battery remaining capacity

Parameter: none

Return value: percentage of battery remaining capacity (1~100)

➤ `int16_t get_bat_charging(void)`

Function: inquire whether the battery is in charging state

Parameter: none

Return value: 0: not in charging state

1: in charging state

➤ `int16_t get_bat_mvolt(void)`

Function: acquire real-time voltage of battery

Parameter: none

Return value: voltage value, unit mV

➤ `int16_t get_bat_mcurrent(void)`

Function: acquire real-time current of battery

Parameter: none

Return value: current value, unit mA

➤ `int16_t get_bat_temp(void)`

Function: acquire battery temperature

Parameter: none

Return value: temperature value, unit degree Celsius

➤ `int16_t get_chassis_work_model(void)`

Function: acquire working state of chassis motor

Parameter: none

Return value: 1: motor in augmentation;

0: motor not in augmentation

➤ `int16_t get_chassis_load_state(void)`

Function: acquire setting value of chassis based on controlling parameter of different loading

Parameter: none

Return value: 0: no-load control parameter;

1: full load controlling parameter

➤ `int16_t get_chassis_mode(void)`

Function: acquire working mode of chassis finite state machine (FSM)

Parameter: none

Return value: 0 locking mode;

1 vehicle control mode;

2 pushing mode;

3 emergency stop mode;

4 error mode

➤ `int16_t get_ctrl_cmd_src(void)`

Function: acquire command origin of motor chassis control

Parameter: none

Return value: 0: control vehicle with remote controller;

1: control vehicle with host computer

➤ `int16_t get_vehicle_meter(void)`

Function: acquire the mileage since the chassis is power up

Parameter: none

Return value: mileage value, unit meter

➤ `uint16_t get_host_version(void)`

Function: acquire the host computer software version

Parameter: none

Return value: host computer software version number

➤ `uint16_t get_chassis_central_version (void)`

Function: acquire the central board firmware version

Parameter: none

Return value: the central board firmware version number

➤ `uint16_t get_chassis_motor_version (void)`

Function: acquire the motor board firmware version

Parameter: none

Return value: the motor board firmware version number

➤ `int16_t get_line_forward_max_vel_fb (void)`

Function: acquire the forward speed limiting feedback value of the chassis

Parameter: None

Return value: the forward speed limiting feedback value of the chassis

➤ `int16_t get_line_backward_max_vel_fb (void)`

Function: acquire the backward speed limiting feedback value of the chassis

Parameter: None

Return value: the backward speed limiting feedback value of the chassis

➤ `int16_t get_angular_max_vel_fb (void)`

Function: acquire the angular speed limiting feedback value of the chassis

Parameter: None

Return value: the angular speed limiting feedback value of the chassis

➤ `int16_t getlapProgress (void)`

Function: Get the progress of IAP upgrades

Parameter: None

Return value:

-1: IAP upgrade failed

0: IAP upgrades are idle or started or interrupted

100: IAP upgrade completed

Other: Percentage of IAP upgrade progress

➤ `void iapCentralBoard (void)`

Function: IAP upgrade of the central board firmware of the chassis

Parameter: None

Return value: none

Note: You need to place the central board firmware "central.bin" in the path of "/sdcard/firmware/" in advance.

➤ `void iapMotorBoard (motor_index_e motor_index )`

Function: IAP upgrade of the motor board firmware of the chassis

Parameter: Parameter is one of the following enumerated values:

Motor_front: Represents the front wheel circuit board

Motor_rear: Represents the rear wheel circuit board

Return value: none

Note: You need to place the motor board firmware "motor.bin" in the path of "/sdcard/firmware/" in advance.

➤ `bool isHostlapOver (void)`

Function: Query if the IAP upgrade process has ended

Parameter: None

Return value: true: the IAP completes or fails or is interrupted

False: IAP not started or in progress

➤ `Int16_t getHostlapResult (void)`

Function: acquire the reason for the end of IAP

Parameter: None

Return value: 3: IAP completes

4: IAP fails

5: IAP is interrupted

Others: IAP not started or in progress

➤ `Int16_t getHostlapErrorCode (void)`

Function: Gets the error code for IAP failure

Parameter: None

Return value: the error code for IAP failure

➤ `int16_t get_calibrate_mid_value_status(void)`

Function: Query if the median headway Angle has been calibrated

Parameter: None

Return value: 1: calibrated;

0: no calibrated.

➤ `void set_cmd_vel(double linearx, double angularz)`

Function: set up the command value of chassis target speed, which needs to be regular transmit once the chassis is enabled. It will be determined as communication failure if the chassis can't receive the command value in continuous 150ms in controlling mode.

Parameter: linear_x: linear velocity command value, unit m/s;

angular_z: angular velocity command value, unit rad/s

Return value: none

➤ `void set_line_forward_max_vel(double linearforwardmax_x)`

Function: set up the max forward linear velocity value of chassis.

Parameter: linearforwardmax_x: max forward linear velocity value of chassis, unit m/s, range 0 - 2.3

Return value: none

➤ `void set_line_backward_max_vel(double linearbackwardmax_x)`

Function: set up the max backward linear velocity value of chassis.

Parameter: linearbackwardmax_x: max backward linear velocity value of chassis, unit m/s, range -0.85 - 0

Return value: none

➤ `void set_angular_max_vel(double angularmax_z)`

Function: set up the max angular velocity command value of chassis.

Parameter: angularmaxz: the max angular velocity command value, unit rad/s, range 0 - 2

Return value: none

➤ `void set_enable_ctrl(uint16_t enableflag)`

Function: set up to enable the chassis to control the vehicle.

Parameter: enable_flag:

1 enable the vehicle control;

0 exit the vehicle control

Return value: none

➤ `void set_smart_car_serial(const char * serialno)`

Function: set up the terminal name of serial port of host computer, e.g. ttyUSB0.

Parameter: serial_no: terminal name of serial port, under the path /dev/ by default, e.g. "ttyUSB0"

Return value: none

➤ `void set_comu_interface (comu_choice_e comu_choice)`

Function: Set up the communication interface between the host computer and the chassis, including serial communication and CAN communication

Parameter: *comu_choice*:

‘comu_serial’ Use a serial port for communication

‘comu_can’ Use a CAN port for communication

Return value: none

➤ `void set_chassis_load_state(int16_t newLoadSet)`

Function: set up the parameter of chassis control based on the different chassis load.

Parameter: newLoadSet:

0: no-load parameter;

1: full load parameter

Return value: none

➤ `void set_chassis_poweroff (void)`

Function: chassis power off controlled by host computer.

Parameter: none

Return value: none

➤ `void setHostlapCanceled (void)`

Function: Interrupt the IAP upgrade process.

Parameter: none

Return value: none

➤ `uint8_t set_calibrate_mid_value(void)`

Function: Sets the command to calibrate the median Angle of the front wheel of the chassis.

Parameter : none.

Return value: 0: successfully set; other: setup failed.

➤ `uint8_t reset_host_power_time_s(uint16_t reset_time_s);`

Function: Set the reset time after power failure of the upper machine

Parameter : `reset_time_s`: reset interval time.

Range: 0~65535. unit: second.

Return value: 0: successfully; other: failed.